

Lecture Note 2: Sampling, Statistics and Sampling Distributions

14.380: Statistical Methods in Economics, Fall 2026

1 Basic Concepts

1.1 Population and Data

In statistics, we usually begin with a *data set* that we model as a realization x from some random vector $\mathcal{X}$. We often refer to the distribution $F_{\mathcal{X}}$ of the data as the *population*.

In econometrics, we typically encounter three types of data. First, we encounter *cross-sectional data*, which we typically model as a set of *independent and identically distributed* random vectors $\mathcal{X} = (X_1, \dots, X_n)$.

Definition 1. Let $\mathcal{X} = (X_1, \dots, X_n)$ be random variables with cdfs $F_1, \dots, F_n$ respectively. Let $\mathcal{F}_A$ denote the joint cdf of the random variables with indices $A \subseteq \{1, \dots, n\}$. $X_1, \dots, X_n$ are *independent and identically distributed* if they satisfy

- i. Mutual independence: for all $A \subseteq \{1, \dots, n\}$, $F_A((x_i)_{i \in A}) = \prod_{i \in A} F_i(x_i)$;
- ii. Identically distributed: for all $i, j = 1, \dots, n$, $F_i(x) = F_j(x)$.

For much of this class, we will focus on the case in which we model our data as being realizations of independent and identically distributed random vectors. Much of what we discuss will generalize to richer settings. For example, in econometrics, we often also encounter time series data X_t for $t = 1, \dots, T$, which we typically model as having dependence between consecutive observations and our data as realizations from some stochastic process. We also often encounter panel data $\mathcal{X} = \{X_{it} : i = 1, \dots, n; t = 1, \dots, T\}$, which we typically model as n independent draws from identically distributed dynamic processes. In other courses, you will study these other settings.

1.2 Parameters and Statistics

The object of interest here is usually some functional of the unknown distribution F_X . Any such function is known as a *parameter*. Formally, a *parameter* is some function $T : \mathcal{F} \rightarrow \mathbb{R}$, where $\mathcal{F}$ is the set of possible CDFs.

In the case of cross-sectional data, it is usually some function of the distribution of one observation F . As an example, we can write the expectation of $X \sim F(\cdot)$ as the functional

$$T(F) = \int x dF(x).$$

We can analogously write the variance as depending on two functionals

$$\int x^2 dF(x) - \left(\int x dF(x) \right)^2.$$

A remarkably large class of estimands can be written in this manner.

The goal of Statistics generally is to render some judgement about a parameter (or population F_X) based on a single draw from this population. This is called *inference*. We will see three types of inference: estimation, confidence set construction and testing. In performing each task we will sometimes make mistakes, and the quality of the procedure will be related to minimizing the size and/or probability of mistakes.

We refer to any function of a random sample as a *statistic*. Thus, $Y = g(\mathcal{X}) = g(X_1, \dots, X_n)$ is a statistic. By construction, it is a random variable. When calculated for our specific data set $y = g(x) = g(x_1, \dots, x_n)$ it produces a single realization of this random variable. The distribution of a statistic is called the *sampling distribution*.

Example 1. Assume you want to figure out whether the penny you have is a fair coin by flipping it $n = 10$ times and recording 0 for each tail and 1 for each head. In this case your data $x = (x_1, \dots, x_{10})$ is a sequence of 0s and 1s of the length 10, where x_i is the result of i -th experiment. This is a single realization of random vector $X = (X_1, \dots, X_{10})$, where $X_i \sim i.i.d. Bernoulli(p)$. The population here is described as $P(x_1, \dots, x_{10}) = \prod_{i=1}^{10} p^{x_i} (1-p)^{1-x_i}$, and is known up to parameter p . So, the only goal is to make some judgement about p . Remember, a statistic just refers to any function of the data set. We may consider many different functions: say,

$$Y_1 = \{\text{number of heads}\} = \sum_{i=1}^{10} X_i,$$

$$Y_2 = \{\text{the order number of the first experiment resulting in heads, with 0 if no heads}\} \\ = X_1 + 2(1 - X_1)X_2 + 3(1 - X_1)(1 - X_2)X_3 + \dots,$$

$$Y_3 = \{\text{the difference b/w \# of heads in the first 6 flips and \# of heads in the remaining 4}\} \\ = \sum_{i=1}^6 X_i - \sum_{i=7}^{10} X_i.$$

If one has data $= (0, 1, 1, 1, 0, 0, 1, 1, 0, 1)$, then $y_1 = 6, y_2 = 2, y_3 = 0$. All statistics Y_1, Y_2, Y_3 are random variables with a distribution depending on p .

Example 2. Assume you want to estimate the average income of a man aged between 25 and 65, who resides in Massachusetts. You were able to get a random sample of income for n such men. Your data is represented by random vector $X = (X_1, \dots, X_n)$, where $X_i \sim F(\cdot)$ i.i.d., where $F(\cdot)$ is the unknown distribution of incomes. The population is described by $F_X(x_1, \dots, x_n) = \prod_{i=1}^n F(x_i)$. The parameter of interest is $\mu = \int u dF(u)$ - the mean of the unknown income distribution. You have one realization of this data set $x = (x_1, \dots, x_n)$, statistic is any function of the data set. Some

examples:

$$Y_1 = \frac{1}{n} \sum_{i=1}^n X_i (\text{average})$$

$$Y_2 = (n/2) - \text{th highest value among } (X_1, \dots, X_n) (\text{median})$$

$$Y_3 = \text{average of 80\% of middle values (drop 10\% smallest and 10\% largest values)} (\text{a trimmed mean})$$

Remark 1. Many statistics are constructed as natural estimators of a particular parameter. As an example, recall from Lecture Note 1 that the p -th quantile of a distribution F is $q_p = \inf\{x: F(x) \geq p\}$. A natural estimator replaces the unknown cdf F with the *empirical cdf* $\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I\{X_i \leq x\}$, which is itself a statistic for each fixed x , yielding the *plug-in* estimator of the p -th quantile

$$\hat{q}_p = \inf\{x: \hat{F}_n(x) \geq p\}.$$

For $p = 1/2$, this delivers (a version of) the sample median Y_2 in the example above. We will study the empirical cdf $\hat{F}_n$ in detail in Lecture Note 4.

1.3 Connection to Random Sampling from a Finite Population

Before continuing, we briefly discuss a canonical motivation for the i.i.d. assumption: modeling our data as a *random sample* from some population.

More concretely, suppose we have a *finite population* U of N units indexed $i = 1, 2, \dots, N$ and each unit is associated with some characteristic x_i . Let us consider a random process that selects one unit from U uniformly at random: that is, we define the discrete random variable X that takes values $\{x_1, \dots, x_N\}$ with equal probability $1/N$. The probability mass function of this random variable is then

$$f(x) = \frac{1}{N} \sum_{i=1}^N I\{x_i = x\}.$$

The probability mass function of X summarizes the fraction of units in the population with characteristic equal to x . Notice that (i) the size of the finite population, and (ii) the support of x_i across units determine the distribution of X . The expectation and variance of X are equivalent to the average of the characteristics in the finite population and the average squared deviation from the mean of all of the units (can you show this?).

Now suppose we wanted to sample n units from the finite population U . It is natural to randomly sample units without replacement, meaning that we only observe each unit once and so each draw affects the remaining population. In this case, we would model our random sample as $(X_1, \dots, X_n)$, but these would not be independent nor identically distributed.

A natural intuition is that if the finite population is sufficiently large, then removing a single unit or a few units changes the distribution of the remaining units by a negligible amount. For example, suppose we are surveying a city with a population of 1 million people, and 500,000 are male and 500,000 are female. We sample n individuals at random without replacement, and let

$X_i = 0$ if individual i is male and $X_i = 1$ if individual i is female. Then, it is easy to see that $P(X_2 = 1) = 0.5$ and $P(X_2 = 1 \mid X_1 = 1) \approx 0.5$. This intuition can be formalized to show that sampling units at random from a hypothetical, infinite *superpopulation* yields a sample that can be treated as-if it were i.i.d. (e.g., Diaconis & Freedman 1980). We make this discussion precise in Section 4, where we derive the sampling distribution of the sample mean under random sampling from a finite population and introduce estimators tailored to more general sampling designs.

1.4 The Sample Mean and Sample Variance

The two most commonly used statistics are the *sample mean*

$$\bar{X}_n = \sum_{i=1}^n X_i/n$$

and the *sample variance*

$$s^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2/(n-1).$$

These statistics have attractive properties as described in the lemma below:

Lemma 1. *If $X_1, \dots, X_n \sim F$ i.i.d. is a random sample of size n from population distribution with mean $\mu = \mathbb{E}X_i$ and variance $\sigma^2 = \text{Var}(X_i)$, then $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathbb{E}[s^2] = \sigma^2$.*

Proof. By linearity of expectation,

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\sum_{i=1}^n X_i/n\right] = \sum_{i=1}^n \mathbb{E}[X_i]/n = \sum_{i=1}^n \mu/n = \mu.$$

To show the second part of the lemma, denote $Y_i = X_i - \mu$ and $\bar{Y}_n = \sum_{i=1}^n Y_i/n$. Note that

$\mathbb{E}[Y_i] = 0$. Thus, $\mathbb{E}[Y_i^2] = V(Y_i) = V(X_i) = \sigma^2$ and $V(\bar{Y}_n) = \sigma^2/n$. Then

$$\begin{aligned}
\mathbb{E}[s^2] &= \mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^n ((X_i - \mu) - (\bar{X}_n - \mu))^2 / (n-1)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^n (Y_i - \bar{Y}_n)^2 / (n-1)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^n (Y_i^2 - 2Y_i\bar{Y}_n + \bar{Y}_n^2) / (n-1)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^n Y_i^2 - 2n\bar{Y}_n^2 + n\bar{Y}_n^2 / (n-1)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^n Y_i^2 - n\bar{Y}_n^2 / (n-1)\right] \\
&= \left(\sum_{i=1}^n \mathbb{E}[Y_i^2] - n\mathbb{E}[\bar{Y}_n^2]\right) / (n-1) \\
&= (n\sigma^2 - \sigma^2) / (n-1) \\
&= \sigma^2.
\end{aligned}$$

□

Remark 2. Why do we divide by $n-1$ rather than n in the sample variance? At first glance, it may seem more natural to estimate the variance $\sigma^2 = \mathbb{E}[(X_i - \mathbb{E}[X_i])^2]$ by replacing the population mean with the sample mean and averaging over the n observations, arriving at the plug-in estimator

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

This estimator does not coincide with the sample variance: $\hat{\sigma}^2 = \frac{n-1}{n} s^2$. By the lemma above, it follows that

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \mathbb{E}[s^2] = \frac{n-1}{n} \sigma^2 \neq \sigma^2,$$

and so the plug-in estimator is biased downward for the population variance. The reason we use $n-1$ instead of n in the denominator of the sample variance is precisely to remove this bias, i.e., $\mathbb{E}[s^2] = \sigma^2$.

2 Calculating the Distribution of a Statistic

In order to make inferences, we often will need to know the distribution of different statistics. There are several ways of getting them.

2.1 Exact distribution

In rare cases, one can actually calculate the exact distribution of a statistic.

Example 3. Consider our earlier example on coin flipping and consider the statistic $Y_1 = \sum_{i=1}^n X_i$. It takes integer values between 0 and 10 with

$$P\{Y_1 = k\} = \frac{10!}{k!(10-k)!} p^k (1-p)^{10-k}.$$

The distribution depends on an unknown p ; and once we know or postulate p , we can calculate the exact distribution of Y_1 .

Example 4. Consider our earlier example on estimating average income. Let us try to figure out the distribution of $Y_1 = \frac{1}{n} \sum_{i=1}^n X_i$. If $X_i \sim F(\cdot)$ with pdf f then $X_1 + X_2 \sim f_2(u)$ where $f_2(u) = \int f(x)f(u-x)dx$. So, Y_1 has pdf

$$f_Y(y) = n \int \dots \int f(ny - y_1 - \dots - y_{n-1}) f(y_1) \dots f(y_{n-1}) dy_1 \dots dy_{n-1},$$

which is a complicated expression and is not very helpful in most situations. It depends in a significant way on the unknown distribution $F(\cdot)$. If we make some strong assumptions about the distribution of our data we may end up with an exact distribution of some statistics. For example, assume $X_i \sim i.i.d.N(\mu, \sigma^2)$, then $Y_1 \sim N(\mu, \sigma^2/n)$. However, this is a strange assumption to make about income distribution (why?).

2.2 Monte-Carlo Method.

Let us consider our earlier example on estimating average income, but now with an alternative assumption of the log-normal distribution. Assume that X_i has the following pdf:

$$f(x; \mu, \sigma^2, \gamma) = (2\pi\sigma^2)^{-1/2} (x - \gamma)^{-1} e^{-\frac{(\log(x-\gamma)-\mu)^2}{2\sigma^2}} I\{x > \gamma\}.$$

Apparently, getting a (closed-form) exact distribution of $Y_1 = \frac{1}{n} \sum_{i=1}^n X_i$ is not that easy. But we can do this numerically by the use of Monte-Carlo method. A typical algorithm for given (μ, σ^2, γ) :

- For $b = 1, \dots, B$, simulate $\mathcal{X}_b^* = (X_{1b}^*, \dots, X_{nb}^*)$, where the X_{ib}^* are independently drawn from $f(x; \mu, \sigma^2, \gamma)$;
- Calculate $Y_b^* = g(\mathcal{X}_b^*)$, in our case $\frac{1}{n} \sum_{i=1}^n X_{ib}^*$;
- If you are interested in:
 - cdf of Y , then $F_Y(s) \approx \frac{1}{B} \sum_{b=1}^B I\{Y_b^* \leq s\}$;
 - probability Y gets into the set A : $P\{Y \in A\} \approx \frac{1}{B} \sum_{b=1}^B I\{Y_b^* \in A\}$;
 - α -quantile of Y : $q_Y(\alpha) \approx Y_{([\alpha B])}^*$, here $(\cdot)$ - stands for the order statistics;

- mean of Y : $\mathbb{E}Y \approx \frac{1}{B} \sum_{b=1}^B Y_b^*$;
- variance of Y : $V(Y) \approx \frac{1}{B-1} \sum_{b=1}^B \left(Y_b^* - \frac{1}{B} \sum_{s=1}^B Y_s^* \right)^2$.

Here all the Y_b^* are i.i.d. from the correct distribution. We control the accuracy here: larger B leads to better accuracy; we are bounded only by power of our computers.

The important assumption here is we assume that we know (or postulate) the distribution of the data.

2.3 Asymptotic approximation

Exact finite-sample distributions of many statistics have complicated forms and cannot be calculated directly if we do not know (and do not want to assume) the distribution of the data. But at the same time that often may be well-approximated if the sample size is large. This is known as asymptotic approximation and relies on CLT, delta-methods, Slutsky theorem and alike. Much of the class will be devoted to developing this framework.

Example 5. In our example of estimating average income, if we are unwilling to assume the distribution of X_i at all, but are willing to assume that it has finite variance, then we can invoke a Central Limit Theorem to argue:

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}X_i \right) \xrightarrow{d} N(0, V(X_i)).$$

Thus the distribution of $Y_1 = \frac{1}{n} \sum_{i=1}^n X_i$ is approximately gaussian with mean $\mathbb{E}X_i$ and variance $V(X_i)/n$. We do not control the quality of this approximation and cannot improve it, but as the sample size grows the approximation should become more accurate.

Asymptotic approximation is probably the most often used way of figuring out the distribution of a statistic in econometrics.

3 Example: Normal Distribution

In practice, it is sometimes assumed that the class of possible distributions form some parametric family. In other words, it is assumed that $F = F(\theta)$ with $\theta \in \Theta$ where Θ is some finite-dimensional set. Then θ is called a parameter and Θ is a parameter space. In this case the cdf F and the corresponding pdf f are often denoted by $F(x|\theta)$ and $f(x|\theta)$. If $X_1, \dots, X_n$ is a random sample from a distribution with pdf $f(x|\theta)$, then joint pdf $f(x_1, \dots, x_n|\theta) = \prod_{i=1}^n f(x_i|\theta)$. For fixed $x_1, \dots, x_n$, $f(x_1, \dots, x_n|\theta)$ as a function of θ is called the *likelihood function*.

One of the most important parametric families is a normal family when $\theta = (\mu, \sigma^2)$ and the population distribution is $N(\mu, \sigma^2)$. The following theorem gives some basic facts about the sample mean and the sample variance for random sample from normal distribution:

Theorem 1. If $X_1, \dots, X_n$ are iid random variables with $N(\mu, \sigma^2)$ distribution, then (1) $\bar{X}_n \sim N(\mu, \sigma^2/n)$; (2) $(n-1)s^2/\sigma^2 \sim \chi_{n-1}^2$; (3) $\bar{X}_n$ and s^2 are independent; and (4) $\frac{\bar{X}_n - \mu}{\sqrt{s^2/n}} \sim t_{n-1}$.

Proof. Let $Z = \bar{X}_n$, $Y_1 = X_1 - \bar{X}_n$, $Y_2 = X_2 - \bar{X}_n$, ..., $Y_n = X_n - \bar{X}_n$. Then $Z, Y_1, \dots, Y_n$ are jointly normal. Obviously, $\mathbb{E}[Z] = \mu$ and $\mathbb{E}[Y_i] = \mu - \mu = 0$ for all $i = 1, \dots, n$. In addition, $V(Z) = \sigma^2/n$. Thus statement (1) holds.

To prove (2), observe that

$$\sum_{i=1}^n (X_i - \bar{X}_n)^2 = Y'_{1:n} Y_{1:n},$$

where $X_{1:n} = (X_1, \dots, X_n)'$ and $Y_{1:n} = (Y_1, \dots, Y_n)'$. We can rewrite this as

$$Y_{1:n} = X_{1:n} - \ell \bar{X}_n = M X_{1:n} \text{ for } M = I - \ell(\ell' \ell)^{-1} \ell',$$

where ℓ is the n -vector of ones, and so

$$(n-1)s^2/\sigma^2 = Y'_{1:n} Y_{1:n}/\sigma^2 = (X_{1:n}/\sigma)' M (X_{1:n}/\sigma) \sim \chi^2_{n-1},$$

since M is an idempotent matrix with $\text{rank}(M) = n-1$ and by our results for quadratic forms of normal random variables.

For any $j = 1, \dots, n$,

$$\begin{aligned} \text{Cov}(Z, Y_j) &= \text{Cov}(\bar{X}_n, X_j - \bar{X}_n) \\ &= \text{Cov}(\bar{X}_n, X_j) - V(\bar{X}_n) \\ &= \sigma^2/n - \sigma^2/n \\ &= 0 \end{aligned}$$

Since uncorrelated jointly normal random variables are independent, we conclude that Z is independent of $Y_1, Y_2, \dots, Y_n$. Moreover, $s^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2/(n-1) = \sum_{i=1}^n Y_i^2/(n-1)$ and statement (3) holds since any functions of independent random variables are independent as well.

Statement (4) follows from statements (1)-(3) and the definition of the t -distribution with $n-1$ degrees of freedom. $\square$

By definition, $t = (\bar{X}_n - \mu)/(s/\sqrt{n})$ is called the t -statistic. Statement (4) in the theorem therefore establishes that the t -statistic has a t -distribution if $X_1, \dots, X_n$ is a random sample from $N(\mu, \sigma^2)$.

Finally, let $X_1, \dots, X_n$ be a random sample from $N(\mu_x, \sigma_x^2)$ and $Y_1, \dots, Y_m$ be a random sample from $N(\mu_y, \sigma_y^2)$. Assume that $X_1, \dots, X_n$ are independent of $Y_1, \dots, Y_m$. Then $F = (s_x^2/s_y^2)/(\sigma_x^2/\sigma_y^2)$ is called an F -statistic. Using the theorem above,

$$F = \frac{s_x^2/s_y^2}{\sigma_x^2/\sigma_y^2} \sim \frac{\chi^2_{n-1}/(n-1)}{\chi^2_{m-1}/(m-1)} = F_{n-1, m-1}$$

Thus, the F -statistic has the F -distribution with $(n-1, m-1)$ degrees of freedom.

4 Random Sampling from a Finite Population, Revisited

In Section 1.3, we motivated the i.i.d. model as an approximation to random sampling from a large finite population, and we appealed to intuition for why the approximation should be accurate. We now make that discussion precise. Recall the setting: a finite population U of N units indexed $i = 1, \dots, N$, each associated with a fixed characteristic x_i , from which we sample n units at random without replacement.

It is convenient to change perspective. Rather than tracking the draws $(X_1, \dots, X_n)$ in order, we track *which units are sampled*. For each unit $i = 1, \dots, N$, define the *sampling indicator* $R_i = 1$ if unit i is included in the sample and $R_i = 0$ otherwise. This change of perspective clarifies what is random and what is fixed: the characteristics $x_1, \dots, x_N$ are a list of fixed numbers, and all randomness comes from the indicators $(R_1, \dots, R_N)$, whose distribution is chosen by the researcher through the *sampling design*. The distribution of any statistic over repeated draws of the sampling design is called its *sampling distribution*, and here the researcher knows this distribution exactly because the researcher designed it. Our estimand is the finite-population mean

$$\bar{x}_N = \frac{1}{N} \sum_{i=1}^N x_i,$$

and it will also be useful to define the finite-population variance $S^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x}_N)^2$.

Consider first *simple random sampling*: every subset of n distinct units is equally likely to be drawn. By symmetry, each unit is included with probability $\mathbb{E}[R_i] = \frac{n}{N}$, and each pair of distinct units is included together with probability $\mathbb{E}[R_i R_j] = \frac{n(n-1)}{N(N-1)}$. The sample mean can be written as

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^N R_i x_i,$$

and so its behavior over the sampling distribution reduces to moment calculations on the indicators.

Theorem 2. *Under simple random sampling of n units from the finite population U ,*

$$\mathbb{E}[\bar{X}_n] = \bar{x}_N \quad \text{and} \quad V(\bar{X}_n) = \left(1 - \frac{n}{N}\right) \frac{S^2}{n}.$$

Proof. Unbiasedness is immediate: $\mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^N \mathbb{E}[R_i] x_i = \frac{1}{n} \sum_{i=1}^N \frac{n}{N} x_i = \bar{x}_N$. For the variance, write

$$V(\bar{X}_n) = \frac{1}{n^2} \left(\sum_{i=1}^N V(R_i) x_i^2 + \sum_{i \neq j} \text{Cov}(R_i, R_j) x_i x_j \right),$$

where $V(R_i) = \frac{n}{N} \left(1 - \frac{n}{N}\right)$ and, for $i \neq j$,

$$\text{Cov}(R_i, R_j) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)} < 0.$$

Substituting these expressions, using the identity $\sum_{i \neq j} x_i x_j = \left(\sum_{i=1}^N x_i\right)^2 - \sum_{i=1}^N x_i^2$, and collecting terms yields

$$V(\bar{X}_n) = \frac{N-n}{nN(N-1)} \left(\sum_{i=1}^N x_i^2 - N\bar{x}_N^2 \right) = \left(1 - \frac{n}{N}\right) \frac{S^2}{n},$$

where the final equality uses $S^2 = \frac{1}{N-1} \left(\sum_{i=1}^N x_i^2 - N\bar{x}_N^2 \right)$. $\square$

The sampling indicators are *negatively* correlated — including unit i makes it less likely that unit j is included — and this negative dependence makes the sample mean more precise than under i.i.d. sampling. The factor $\left(1 - \frac{n}{N}\right)$ is called the *finite population correction*, and it delivers the promised intuition as two limiting cases. First, if $n = N$, the researcher has conducted a census: every unit is observed, $\bar{X}_n = \bar{x}_N$ exactly, and the variance is zero — there is no sampling uncertainty left. Second, as $N \rightarrow \infty$ with the sampling fraction $\frac{n}{N} \rightarrow 0$, the finite population correction tends to one and $V(\bar{X}_n) \rightarrow \frac{S^2}{n}$, which is exactly the variance of the sample mean under i.i.d. draws from the distribution that places mass $\frac{1}{N}$ on each x_i . In this sense, the i.i.d. model of Section 1 is the large-population limit of random sampling without replacement — we imagine our data as drawn from a hypothetical, infinite *superpopulation*. This as-if-i.i.d. approximation can be formalized quite generally (e.g., Diaconis & Freedman 1980); Aronow and Miller develop the finite-population perspective in detail.

Simple random sampling gives every unit the same chance of selection, but real surveys often do not. The researcher may deliberately oversample a small subgroup to measure it precisely, or sample firms with probability proportional to their size. Unequal selection probabilities break the sample mean: units that are oversampled are overrepresented, and the sample mean is biased for $\bar{x}_N$. Can the researcher still estimate $\bar{x}_N$ without bias? Yes — provided the selection probabilities are known, which they are when the researcher designed the survey. Suppose the sampling design includes unit i with known probability $\pi_i = \mathbb{E}[R_i] > 0$ for every i , and let $\pi_{ij} = \mathbb{E}[R_i R_j]$ denote the probability that units i and j are included together (with $\pi_{ii} = \pi_i$). The *Horvitz-Thompson estimator* weights each sampled unit by the inverse of its inclusion probability:

$$\hat{\mu}_{HT} = \frac{1}{N} \sum_{i=1}^N R_i \frac{x_i}{\pi_i}.$$

Intuitively, a unit sampled with probability π_i “stands in” for $\frac{1}{\pi_i}$ units of the population, and the weighting undoes the distortion that the design built in.

Theorem 3 (Horvitz-Thompson). *For any sampling design with known inclusion probabilities $\pi_i > 0$ for all i ,*

$$\mathbb{E}[\hat{\mu}_{HT}] = \bar{x}_N \quad \text{and} \quad V(\hat{\mu}_{HT}) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N (\pi_{ij} - \pi_i \pi_j) \frac{x_i x_j}{\pi_i \pi_j}.$$

Proof. Unbiasedness takes one line:

$$\mathbb{E}[\hat{\mu}_{HT}] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[R_i] \frac{x_i}{\pi_i} = \frac{1}{N} \sum_{i=1}^N \pi_i \frac{x_i}{\pi_i} = \bar{x}_N.$$

For the variance, $\hat{\mu}_{HT}$ is a linear combination of the indicators R_i with fixed coefficients $\frac{x_i}{N\pi_i}$, and so

$$V(\hat{\mu}_{HT}) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \text{Cov}(R_i, R_j) \frac{x_i}{\pi_i} \frac{x_j}{\pi_j},$$

where $\text{Cov}(R_i, R_j) = \mathbb{E}[R_i R_j] - \mathbb{E}[R_i] \mathbb{E}[R_j] = \pi_{ij} - \pi_i \pi_j$ (and, for $i = j$, $\text{Cov}(R_i, R_i) = \pi_i(1 - \pi_i)$, which is the $\pi_{ii} - \pi_i^2$ term). $\square$

Notice what this theorem asks of the researcher: nothing about the characteristics $x_1, \dots, x_N$, which may be arbitrary fixed numbers, and everything about the design, which must be known. Unbiasedness here is a statement about the sampling distribution alone.

Example 6. The simplest design with unequal probabilities is *Poisson sampling*: each unit is included independently, with $R_i \sim \text{Bernoulli}(\pi_i)$. Independence across units implies $\pi_{ij} = \pi_i \pi_j$ for $i \neq j$, so only the diagonal terms survive and the variance simplifies to

$$V(\hat{\mu}_{HT}) = \frac{1}{N^2} \sum_{i=1}^N \frac{1 - \pi_i}{\pi_i} x_i^2.$$

The formula reveals the cost of small inclusion probabilities: a unit with tiny π_i carries the large weight $\frac{1}{\pi_i}$ on the rare occasions it is sampled, and it contributes to the variance in proportion to $\frac{1}{\pi_i}$. Precise estimation requires that no unit be both important and almost never sampled. As an exercise, verify that simple random sampling is the special case $\pi_i = \frac{n}{N}$ and $\pi_{ij} = \frac{n(n-1)}{N(N-1)}$, and that the general variance formula then reduces to the finite population correction formula derived above.

Remark 3. Since the π_i and π_{ij} are known, the variance can itself be estimated from the sample by weighting each observed term by its inclusion probability — the same inverse-weighting trick applied to the variance formula. The idea of reweighting observations by the inverse of a known (or estimated) probability of observation extends far beyond survey sampling: it reappears throughout econometrics as *inverse probability weighting*, wherever some units are observed more readily than others.